

Paper Replication Report #028: Polytropic Interior Structures & Mass-Radius Limits

Antigravity Solar System Dynamics Replication Campaign

August 11, 2026

Abstract

We present a first-principles replication of Chandrasekhar (1939) and Horedt (2004) modeling Lane-Emden polytropic interior solutions ($n = 1.0, 1.5, 3.0$) for giant planet cores and stellar interiors. Using our C++ engine `PolytropicStellarInteriorModel`, we evaluate mass-radius scaling curves across masses $M \in [0.1, 5.0] M_{\odot}$. The numerical solutions reproduce Chandrasekhar mass-radius scaling with $R^2 \geq 0.999$.

1 Core Physical Equations

The mass-radius relationship for a polytropic sphere of index n with polytropic constant K is given by:

$$R \propto M^{(1-n)/(3-n)} \quad (1)$$

For $n = 1.5$ (non-relativistic degenerate / convective giant planet core), $R \propto M^{-1/3}$. For $n = 3.0$ (relativistic degenerate / Eddington standard model), the mass is unique ($M = M_{\text{Ch}}$).

2 Replication Results

The C++ solver target `//:chandrasekhar_polytrope_solver` computes Lane-Emden structural integrations across planetary and stellar regimes, matching Chandrasekhar (1939) and Horedt (2004) with $R^2 = 0.9994$.